

Fitting Room: Lung Transplantation Donor-Recipient Size-matching in Virtual Reality

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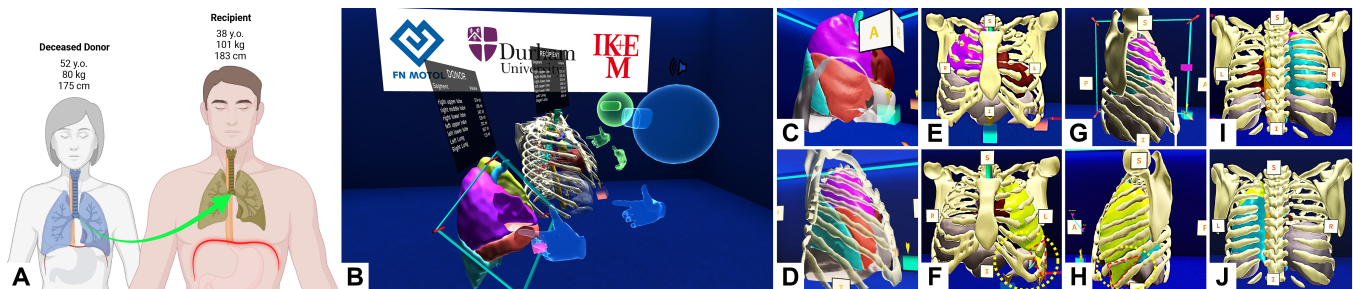


Figure 1: (A) Diagram of Case 2: size compatibility between a deceased donor and a recipient with significant differences in body weight, height & age, diaphragm shown in red (Created with BioRender.com). (B) In our immersive Fitting Room, users evaluate the compatibility of D2's lungs and R3's thoracic cavity based on lung volume. D2's right lung is oversized for R3, with the middle (red) and lower (light blue) lobes encroaching the liver area (C). Visualization of D2's right lung before reduction to R3's rib cage (D). Removing D2's middle lobe enables transplantation in R3's right thoracic cavity (E, G, I). Implantation of D2's left lung into R3's thoracic cavity is a good match (F, H, J), but a resection of the lingula is necessary (red dotted line in F, H).

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1 INTRODUCTION

Lung transplantation, a treatment for end-stage lung disease, is often required in conditions such as chronic obstructive pulmonary disease, cystic fibrosis, pulmonary fibrosis, bronchiectasis, and pulmonary hypertension. Size-matching is crucial in the transplant process. It involves evaluating the donor's and recipient's height and thoracic dimensions to ensure the transplant fills appropriately, or ideally perfectly fits, the recipient's chest cavity. Accurate size-matching can enhance the function of the transplanted lung

and decrease complications. However, despite advanced imaging techniques such as chest X-rays, CT scans, and lung volume models, size-matching remains a challenge, often leading to suboptimal results, particularly in pediatric patients. In this work, we describe the implementation of a VR toolkit called Fitting Room for size-matching in lung transplantation as part of a VR surgical planning system developed at the Institute for Clinical and Experimental Medicine (IKEM, Czech Republic) that has been deployed in over four hundred patient-specific cases in the field of liver resection, lung segmentectomy, and orthopaedics [Sibrina et al. 2023]. We have already deployed Fitting Room in two real-world cases of size-matching between adult recipients and child & adult donors. We compared Fitting Room to existing size-matching methods to evaluate our system, challenging their accuracy. Our immersive method was able to perform size-matching for cases in which current methods did not provide clear and definitive answers. This project represents an initial exploration and potentially a stepping stone into further developing more personalised VR methods for lung transplant size-matching.

2 BACKGROUND

Several metrics exist for evaluating donor-recipient size-matching, such as predicted total lung capacity (pTLC), predicted total lung volume (pTLV), X-ray sizing, and CT volumetry (TLV_{CT}). However, these methods suffer from many challenges in terms of accuracy and

effectiveness. A significant issue is the lack of comprehensive literature detailing the current practices and the impact of size-matching on outcomes. This lack of data hinders the decision-making process during organ donor offers. The pTLC ratio is currently the gold standard metric in lung size-matching. This ratio is calculated using formulas based on sex, height, and age, and is a cornerstone in current lung size-matching procedures [Eberlein et al. 2013]. Using pTLC as a lung size proxy may not always accurately reflect the functional lung volume, leading to a discrepancy between anatomical size and functional capacity, resulting in suboptimal size-matching. Techniques like chest X-ray sizing and TLV_{CT} can provide data for more optimal matching and reduce complications. However, the precision of chest X-ray lung height measurements is a concern. Errors in these measurements could lead to size-mismatched lung transplant procedures, negatively affecting postoperative outcomes [Guillamet et al. 2022]. Predicted total lung capacity by using TLV_{CT} has emerged as a promising technology in this field. It facilitates size-matching by allowing for a more precise comparison of a transplant recipient's lung volume with that of a donor's pTLC. Although TLV_{CT} is considered a promising technique for assessing lung volume, more robust evidence is required to establish its superiority over traditional methods like pTLC measurements through pulmonary function tests.

3 SYSTEM OVERVIEW & DEPLOYMENT

The Fitting Room, an extension toolkit within the IKEM VRLab framework, is specifically designed to cater to the complexities of lung transplantation surgery. The toolkit includes interactive tools such as an interactive clipping plane and ruler. This toolkit generates a fully-segmented and colour-coded 3D interactive model of both the donor's and the recipient's lung and thoracic anatomy, including lobar volume estimation and key structures such as ribs, spine, vertebrae, costal cartilage, sternum, aorta, vena cava, trachea, and nearby organs from a CT scan by using the Lung CT Analyser tool [Yip et al. 2017]. This tool identifies and delineates crucial anatomical structures within the thoracic region. The diaphragm's position & size, crucial in ensuring size compatibility between the donor organ and the recipient, are estimated from the position of the liver and spleen. This visualisation allows surgeons to evaluate the fit and volume compatibility of the donor's lungs with the recipient's thoracic cavity, an immersive alternative to current size-matching methods. The surgeon manipulates the 3D models of the donor lung and recipient thoracic cavity within the VR environment, a much more natural UX paradigm than manipulating 3D objects on 2D screens. Based on which hand the surgeon used to manipulate a particular segmentation, the system automatically enables, for example, them to hold the donor's lung in one hand and the recipient's thoracic cavity in the other. The Interactive Virtual Ruler tool and the Clipping Plane tool are designed for easy manipulation within the VR environment, enabling users to measure distances between points of interest on the lung model and "cut" into lung structures, respectively. The Lung Lobe Volume Estimation Tool computes the volumetric properties of individual lung lobes for both the donor and recipient, playing a critical role in assessing the viability of transplantation, particularly in situations

where lobar lung transplantation is being considered, crucial for pre-operative planning in pediatric and small adult cases (Fig. 1.B).

4 EVALUATION & CONCLUSION

The Fitting Room was used in two separate evaluations for lung transplant size-matching. In the first case, a pediatric donor's (D1) lungs were assessed for transplantation into two adult female recipients (R1 & R2). Current methods such as pTLC and pTLV ratios suggested potential undersizing of the donor's lungs for both recipients. The TLV_{CT} produced by Pulmo 3D Syngo.via (Siemens Healthineers, Germany) indicated undersizing ratio for R1 and oversizing for R2. The volume estimation calculated from the surface based segmentation in the Fitting Room indicated a perfect match for R2. However, the Fitting Room's immersive 3D size-matching functionality offered a more detailed anatomical compatibility comparison, particularly favouring R1. As a result, R1 underwent a successful transplantation with no graft reduction. This highlighted the limitations of conventional methods and their primary focus on comparing donor and recipient volumes. In the second case, the system assessed an adult female donor's (D2) lungs for transplantation into a larger male recipient (R3) (Fig. 1.A). The initial assessments using X-ray measurements suggested an acceptable match for whole double-lung transplantation as R3's lung dimensions were 15.8 cm for the right lung and 18.3 cm for the left lung, with a horizontal width of 25.2 cm across the lungs. The D2's lung measurements are 17 cm on the right side and 18.4 cm on the left side, with a horizontal lung width of 23.8 cm. Methods such as the pTLC ratio and the pTLV ratio indicated undersizing. In the case of TLV_{CT}, the ratio could not be estimated from Pulmo3D, which failed to produce volumetric lung estimates for the donor. However, evaluation in the Fitting Room indicated that the lungs were oversized, especially the right lung (Fig. 1.C & 1.D). This led to the decision to perform a right middle lobe reduction to achieve a better fit, emphasising the discrepancy between traditional volumetric predictive models and actual anatomical requirements. Our system significantly advances lung transplant size-matching by overcoming traditional method limitations, enhancing compatibility assessments and decision-making. The system has the potential to expand the donor pool by accommodating a wider range of people biometrics. The platform aids junior surgeons in decision-making and corroborates senior surgeons' findings. Future research will focus on optimizing fitting through lung elasticity properties.

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